Addendum to Paper — Interpreting GFlowNets for Drug Discovery: Extracting Actionable Insights for Medicinal Chemistry

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February 28, 2026

The sparse autoencoder analysis presented in the original paper employs a compressive bottleneck architecture (256→128→256) designed to identify the dominant axes of variation in the embedding space, a form of sparse dimensionality reduction analogous to sparse PCA [7]. This analysis reveals interpretable physicochemical axes such as polarity, size, and lipophilicity, confirming that SynFlowNet organizes its internal representations along chemically meaningful directions.

However, compressive autoencoders and overcomplete dictionary learning answer fundamentally different scientific questions. The compressive approach identifies *which major axes* the model uses; an overcomplete sparse autoencoder [6] instead asks *which atomic features* the model has learned to represent, including those distributed across superposition in the embedding space [5]. The superposition hypothesis [5] posits that neural networks encode more features than they have dimensions, exploiting near-orthogonal directions in high-dimensional space. Recovering these features requires an *overcomplete* dictionary—one in which the number of latent features exceeds the embedding dimension—as established by Bricken et al. [2] and Cunningham et al. [4] in the context of language models.

Motivated by this distinction, we extend our analysis in three ways. First, we train an overcomplete SAE with a 4× expansion factor (256→1024→256) using BatchTopK activation [3], which enforces exact sparsity structurally rather than through ℓ_1 regularization and avoids the coefficient-tuning sensitivity of penalty-based approaches. Second, we replace the multilayer perceptron probes of the original paper which provide an upper bound on information accessibility via nonlinear decoding—with single-layer linear probes that rigorously test *linear* decodability [1]. Third, we evaluate whether individual SAE latent features encode interpretable chemical concepts through probe weight analysis and substructure enrichment, enabling feature-level mechanistic interpretation of the kind demonstrated by Bricken et al. [2] for language models and extended here to a molecular generative policy.

The two analyses are therefore complementary rather than redundant: the compressive SAE characterizes the global structure of the learned representation, while the overcomplete SAE exposes the fine-grained chemical vocabulary encoded within it. Together, they provide a multi-scale account of how SynFlowNet organizes chemical knowledge—from dominant physicochemical axes to context-sensitive, monosemantic feature detectors.

0.1 Overcomplete BatchTopK Sparse Autoencoder

0.1.1 Architecture

Given a frozen graph-level embedding $\mathbf{h} \in \mathbb{R}^{256}$ from the final graph transformer layer, the overcomplete SAE encodes a sparse latent representation $\mathbf{z} \in \mathbb{R}^{1024}$ and reconstructs $\hat{\mathbf{h}} \in \mathbb{R}^{256}$:

$$\mathbf{z}_{\text{pre}} = \text{ReLU}(\mathbf{W}_{\text{enc}}(\mathbf{h} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \tag{1}$$

$$\mathbf{z} = \text{BatchTopK}(\mathbf{z}_{\text{pre}}, k) \tag{2}$$

$$\hat{\mathbf{h}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}} \tag{3}$$

where $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{1024\times 256}$, $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{256\times 1024}$, and $\mathbf{b}_{\text{pre}}$ is initialized to the training set mean. The dictionary size of 1024 represents a 4× expansion over the input dimension, providing sufficient capacity for monosemantic feature discovery while remaining tractable for 30,676 training molecules.

The BatchTopK activation enforces exact sparsity at the batch level: for a batch of B samples, all $B \times 1024$ pre-activations are collected, and a threshold is chosen such that exactly $k \times B$ activations survive (with $k = 64$). Values below the threshold are zeroed out, with gradients passed through via the straight-through estimator. This eliminates the need to tune an ℓ_1 coefficient, as sparsity is enforced structurally.

The training objective is mean squared reconstruction error only:

$$\mathcal{L} = \|\mathbf{h} - \hat{\mathbf{h}}\|_2^2 \tag{4}$$

Decoder columns are renormalized to unit length after each optimizer step to prevent feature magnitude from being absorbed into the decoder.

0.1.2 Dead Neuron Resampling

A known challenge with TopK-family SAEs on moderately sized datasets is that a large fraction of dictionary features may become permanently inactive ("dead neurons"). To address this, we apply a resampling procedure every 10 epochs starting at epoch 20: features that have not fired on any training sample during the preceding epoch are reinitialized. Specifically, the encoder row and decoder column for each dead feature are set to the normalized direction of a randomly sampled high-reconstruction-loss training example, and the corresponding encoder bias is reset to zero. We cap resampling at 2% of features per round to maintain training stability, observing brief NMSE spikes (∼0.04–0.06) that recover within 2–3 epochs.

0.1.3 Training Details

The SAE is trained on frozen SynFlowNet embeddings extracted from the final graph transformer layer after mean pooling over atoms. We use Adam optimization with learning rate 3×10^{-4}, batch size 512, and a 90/10 train/validation split for 50 epochs. Embeddings are standardized to zero mean and unit variance per feature prior to training. All experiments use a fixed random seed of 42.

0.1.4 SAE Evaluation Metrics

Table 1 summarizes the trained SAE's performance.

Table 1: BatchTopK SAE evaluation metrics (dict_size=1024, k=64).

Metric	**Value**
Normalized MSE (NMSE)	0.0008
L_0 (mean active features/sample)	64.0
Feature activation %	6.25%
Dead neurons	54%
Active dictionary features	470 / 1024

The NMSE of 0.0008 indicates near-lossless reconstruction: the SAE preserves >99.9% of the variance in the original embeddings while activating only 64 of 1024 features per molecule (6.25% sparsity). Approximately 470 features are utilized across the dataset, with 54% remaining inactive despite resampling. This dead neuron rate, while elevated relative to SAEs trained on much larger language model activation datasets, reflects the limited diversity of 30,676 molecular embeddings and is sufficient for downstream interpretability analysis.

0.2 Linear Probe Baselines on Raw Embeddings

To establish baselines for information content, we train single-layer linear probes (`nn.Linear`$(256, 1)$) on the raw graph-level embeddings for 11 molecular properties. Four are binary classification tasks (halogen presence, aromaticity, urea substructure, boron presence) evaluated by accuracy and AUROC, and seven are regression tasks (drug-likeness, complexity, lipophilicity, molecular size, polarity, flexibility, and molecular weight) evaluated by R^2.

Unlike the MLP probes in Section 2.3, these single-layer probes can only extract *linearly* represented information, providing a strict test of linear decodability. Training uses Adam optimization (lr=10^{-3}), early stopping with patience 10, and an 80/10/10 train/validation/test split.

Table 2: Linear classification probe results on raw SynFlowNet embeddings.

Probe	Prevalence	Test Accuracy	Test AUROC	Test F1
has_boron	43.7%	99.5%	1.000	0.994
has_halogen	60.3%	97.5%	0.998	0.979
is_aromatic	97.7%	96.6%	0.998	0.983
has_urea	1.7%	73.7%	0.804	0.071

0.2.1 Classification Probe Results

Boron, halogen, and aromatic ring presence are all linearly decodable from SynFlowNet embeddings with AUROC ≥ 0.998 using a single linear layer, confirming that these fundamental structural motifs are represented along accessible directions in embedding space. The urea probe achieves AUROC 0.804 but near-zero F1 due to extreme class imbalance (1.7% prevalence); the embedding captures partial urea signal but a linear probe cannot reliably classify such rare motifs.

0.2.2 Regression Probe Results

Table 3: Linear regression probe results (R^2) on raw SynFlowNet embeddings, compared to SAE factor predictions from Section 2.2.

Property	Test R^2 (Linear Probe)	Test R^2 (SAE Factors, §2.2)	ΔR^2
Size (heavy atom count)	1.000	0.711	+0.289
Complexity (Bertz CT)	0.970	0.750	+0.220
Polarity (TPSA)	0.945	0.918	+0.027
Molecular weight	0.987	—	—
Lipophilicity (Crippen LogP)	0.806	0.664	+0.142
Drug-likeness (QED)	0.756	0.251	+0.505
Flexibility (rotatable bonds)	0.688	0.502	+0.186

Linear probes on the raw 256-dimensional embeddings substantially outperform the compressive SAE factor predictions from Table 1 across all properties. Size is encoded perfectly ($R^2 = 1.000$), which is expected since graph neural networks operate directly on molecular graphs where atom count is trivially available. Complexity ($R^2 = 0.970$) and polarity ($R^2 = 0.945$) are near-perfectly linearly decodable, while drug-likeness ($R^2 = 0.756$)—a nonlinear composite of multiple descriptors—and flexibility ($R^2 = 0.688$) are partially captured. These baselines confirm that SynFlowNet embeddings encode rich physicochemical information in linearly accessible directions.

The improvement over the compressive SAE factors is expected: a 256→128 bottleneck necessarily discards information, particularly for properties like size and drug-likeness that depend on many embedding dimensions simultaneously. The overcomplete SAE analysis below tests whether expanding rather than compressing the representation recovers this information in an interpretable, monosemantic form.

0.3 Phase 2A: Information Preservation Through SAE Reconstruction

To quantify how much chemically relevant information the overcomplete SAE preserves, we pass test-set embeddings through the full encode-decode pipeline ($\mathbf{h} \to \mathbf{z} \to \hat{\mathbf{h}}$) and evaluate the *same pre-trained probes* on the reconstructed embeddings $\hat{\mathbf{h}}$, without retraining. Any degradation in probe performance reflects information lost by the SAE's sparse bottleneck.

Table 4: Probe performance on SAE reconstructions vs. raw embeddings. Δ indicates change from baseline.

Property	**Baseline**	**Reconstruction**	Δ	**% Preserved**
Classification (Accuracy)				
has_halogen	97.5%	94.6%	−2.8%	97.1%
is_aromatic	96.6%	96.6%	+0.0%	100.0%
has_urea	73.7%	74.3%	+0.6%	100.0%
has_boron	99.5%	98.5%	−1.0%	99.0%
Regression (R^2)				
Size	1.000	1.000	−0.000	100.0%
Complexity	0.970	0.939	−0.030	96.8%
Polarity	0.945	0.911	−0.033	96.5%
Mol. weight	0.987	0.960	−0.027	97.3%
Lipophilicity	0.806	0.759	−0.047	94.1%
Drug-likeness	0.756	0.733	−0.023	96.9%
Flexibility	0.688	0.656	−0.032	95.3%

The SAE preserves >94% of probe-detectable information across all properties. Size is perfectly preserved, and most properties degrade by only 0.02–0.05 in R^2. Lipophilicity shows the largest drop ($\Delta R^2 = -0.047$), consistent with LogP's dependence on subtle atom-level electronic effects that may be partially lost during sparse compression. Aromaticity is perfectly preserved, likely because it depends on graph-topological features that the SAE's 64 active features can efficiently represent.

These results demonstrate that the BatchTopK SAE achieves a favorable reconstruction–sparsity tradeoff: only 64 of 1024 features are active per molecule, yet essentially all chemically relevant information is retained.

0.4 Phase 2B: Chemical Property Prediction from SAE Latent Features

We next train *new* linear probes directly on the 1024-dimensional sparse latent vectors $\mathbf{z}$, testing whether the SAE's learned features individually encode chemical information.

Table 5: Linear probes on SAE latent features ($\mathbf{z} \in \mathbb{R}^{1024}$) compared to baselines on raw embeddings ($\mathbf{h} \in \mathbb{R}^{256}$).

Property	**Baseline (raw h)**	**SAE Latents (z)**	Δ
Classification (Accuracy / AUROC)			
has_halogen	97.5% / 0.998	94.1% / 0.986	−3.4%
is_aromatic	96.6% / 0.998	99.1% / 0.995	+2.5%
has_urea	73.7% / 0.804	74.8% / 0.813	+1.1%
has_boron	99.5% / 1.000	98.6% / 0.998	−0.9%
Regression (R^2)			
Size	1.000	1.000	−0.000
Complexity	0.970	0.948	−0.022
Polarity	0.945	0.915	−0.030
Mol. weight	0.987	0.965	−0.022
Lipophilicity	0.806	0.761	−0.045
Drug-likeness	0.756	0.754	−0.002
Flexibility	0.688	0.685	−0.003

Latent probes nearly match raw embedding baselines, with most R^2 values within 0.03 of the original. Notably, drug-likeness ($\Delta R^2 = -0.002$) and flexibility ($\Delta R^2 = -0.003$) are almost perfectly preserved in the sparse representation, and aromaticity classification *improves* from 96.6% to 99.1%—suggesting the SAE has learned features that make aromatic ring detection more linearly accessible than in the raw embedding.

These results confirm that the overcomplete SAE's sparse features collectively encode the same chemical information as the dense embeddings, enabling the feature-level interpretability analysis that follows.

0.5 Feature-Level Interpretability

The primary advantage of overcomplete SAEs over dense embeddings is that individual latent features can be inspected for semantic meaning. For each of the 11 probe tasks, the linear probe's weight vector $\mathbf{w} \in \mathbb{R}^{1024}$ assigns a scalar importance to each SAE feature. Sorting by $|\mathbf{w}_i|$ identifies which features drive each property prediction.

0.5.1 Feature Sharing and Monosemanticity

Across all 11 probe weight vectors, 118 unique SAE features appear in at least one property's top-20 list. Of these, 65 (55%) are *monosemantic*—appearing in the top-20 for only a single property—while 53 (45%) are *polysemantic*, appearing for two or more properties. This ratio suggests a healthy degree of disentanglement: the majority of important features encode property-specific chemical information, while a minority capture broader structural variation.

0.5.2 Feature Characterization via Substructure Enrichment

To interpret individual features, we compare the 30 highest-activating molecules against 30 randomly sampled silent molecules (activation = 0) for each feature, computing RDKit molecular descriptors and SMARTS-based substructure prevalence in each group. Statistical significance is assessed via Mann-Whitney U tests for descriptors and Fisher's exact test for substructure prevalence, with enrichment defined as a prevalence difference exceeding 30 percentage points ($p < 0.01$).

We selected five features for detailed analysis based on interpretive clarity and complementarity: two monosemantic halogen-subtype detectors (#868, #484), one polysemantic large-scaffold detector (#429), and two supporting features that demonstrate how the SAE factorizes halogen and boron signals (#898, #666). Table 6 summarizes these features alongside two additional features included for completeness.

0.5.3 Halogen Subtype Factorization

The SAE's most striking organizational principle is its factorization of halogen chemistry into distinct, non-overlapping features. Features #868 and #484 illustrate this clearly (Figure 2):

Feature #868 (brominated polyaromatic, no boron) activates for molecules containing bromine in the context of extended aromatic systems: 100% of top-activating molecules contain Br (vs. 20% of silent molecules), with a mean of 4.7 aromatic rings, 5.0 total rings, and enrichment for pyridine (+37%) and aromatic nitrogen (+27%). Critically, boronic acid is *absent* from all top-activating molecules (0% vs. 57%), indicating this feature detects brominated aromatic scaffolds that are *not* boronic acid intermediates.

Feature #484 (fluorinated multi-ring, lipophilic) is the complementary halogen detector: 80% of top-activating molecules contain fluorine (vs. 20% of silent molecules), with high lipophilicity (mean LogP 5.8 vs. 3.1), 3.8 aromatic rings, and low polar surface area (78 Å^2 vs. 100 Å^2). No significant enrichment for bromine or boronic acid is observed.

These two features demonstrate that SynFlowNet does not merely encode "halogen presence" as a single signal. Rather, fluorine and bromine chemistry are represented along distinct axes, capturing their divergent physicochemical effects: fluorine's role in tuning lipophilicity and metabolic stability versus bromine's association with extended aromatic architectures.

Table 6: Interpretable SAE feature characterization. "Active %" is the fraction of molecules activating each feature. Enriched/depleted motifs show prevalence in top-30 activating vs. silent molecules. All reported enrichments are significant at $p < 0.01$.

Feature	Interpretation	Active %	Enriched Motifs	Depleted Motifs
Monosemantic halogen-subtype detectors				
#868	Brominated polyaromatic (no boron)	0.3%	Br (100% vs 20%), pyridine (50% vs 13%)	boronic acid (0% vs 57%), alcohol (3% vs 47%)
#484	Fluorinated multi-ring (lipophilic)	0.9%	F (80% vs 20%)	—
Polysemantic scaffold detector				
#429	Large flexible boronic amine	2.8%	sec-amine (70% vs 30%), boronic acid (87% vs 33%), Br (90% vs 43%), F (70% vs 27%)	alcohol (17% vs 60%), ester (7% vs 43%)
Complementary boron/halogen factorization				
#898	Brominated lipophilic boronic acid	3.1%	Br (100% vs 33%), boronic acid (63% vs 30%)	ether (23% vs 50%)
#666	Flexible boronic acid/ phenol (no halogen)	1.4%	boronic acid (83% vs 47%), phenol (70% vs 33%)	Br (13% vs 43%), F (0% vs 23%)
Additional features (Supplementary)				
#386	Small drug-like boronic alcohol	0.2%	boronic acid (93% vs 53%), alcohol (77% vs 43%)	aromatic_N (53% vs 87%) Br (10% vs 43%)
#721	Boronic acid H-bond acceptor	1.9%	boronic acid (67% vs 33%), phenol (73% vs 33%)	—

0.5.4 Complementary Bromine Detectors

The distinction between Features #868 and #898 further illustrates the SAE's context-sensitivity (Figure 2). Both features activate for brominated molecules (100% Br prevalence in top-30), but they diverge in boron content: #868 excludes boronic acids entirely (0% vs. 57%), while #898 is enriched for boronic acids (63% vs. 30%). Feature #666 completes the picture as the anti-halogen boronic acid detector: boronic acid prevalence is 83% in top-activating molecules, while Br prevalence is only 13% (vs. 43% in silent molecules) and F is entirely absent.

Together, #868, #898, and #666 form a complementary system that factorizes the chemical space along two orthogonal axes: halogenation (present vs. absent) and boron content (boronic acid vs. non-boron scaffold). This factorization is not imposed by the training objective—which contains no explicit chemical supervision—but emerges from the structure of the embedding space itself.

0.5.5 Polysemantic Scaffold Detection

Feature #429 is the most active polysemantic feature (2.8% activation rate) and participates in the top-20 weight lists for five properties: halogen, boron, drug-likeness, flexibility, and molecular weight (Figure 2). Its enrichment profile is correspondingly rich: boronic acid/ester (+53%), Br (+47%), F (+43%), secondary amine (+40%), and pyridine (+43%), with the largest molecular weight gap of any feature (mean 605 Da vs. 445 Da in silent molecules, $p < 10^{-4}$).

This feature functions as a "large elaborated scaffold" detector: it activates for molecules that are simultaneously large, halogenated, boronic, flexible (8.5 rotatable bonds), and amine-containing. Its polysemantic nature is chemically coherent—these properties co-occur in the dataset because large boronic acid building blocks used in synthesis tend to carry multiple functional handles (halogens, amines) and have high molecular weight. Feature #429 captures this co-occurrence structure as a single latent direction.

0.5.6 Decomposition of Drug-Likeness

The feature–property weight heatmap (Figure 1) reveals how the SAE decomposes drug-likeness prediction into interpretable components. Three features from our Tier 1 and 2 analysis carry the largest negative weights for QED, each encoding a distinct structural liability:

- **Feature #484** ($w = -2.65$): excessive complexity and lipophilicity. Top-activating molecules have mean LogP 5.8 and 3.8 aromatic rings—fluorinated multi-ring systems that violate Lipinski's LogP criterion.
- **Feature #429** ($w = -2.46$): excessive size and flexibility. Mean molecular weight of 605 Da and 8.5 rotatable bonds place these molecules beyond both Lipinski's molecular weight cutoff and Veber's flexibility threshold.
- **Feature #898** ($w = -2.16$): excessive lipophilicity driven by halogenation. Mean LogP 4.6 with 100% bromine prevalence, approaching the LogP$\leq$5 boundary.

Additional features with smaller negative weights (see Supplementary Table S1) contribute to the prediction through excessive flexibility and amine enrichment, but the three features above account for the dominant decomposition. Together, they partition the model's implicit understanding of drug-likeness into interpretable structural failure modes—excessive complexity (#484), excessive size (#429), and excessive lipophilicity (#898)—each grounded in a specific chemical context. This decomposition aligns with established medicinal chemistry heuristics (Lipinski's rule-of-five, Veber's oral bioavailability rules) and provides actionable, feature-level rationales for why the model assigns low drug-likeness scores to specific molecular classes.

0.5.7 Boronic Acid Context Specialization

A recurring theme across the analyzed features is the SAE's context-dependent representation of boronic acid chemistry—a dominant motif in this dataset (43.7% of molecules contain boron). Rather than learning a single "boronic acid detector," the SAE distributes boronic acid recognition across multiple features that distinguish boronic acids by their molecular context:

- **#386**: Small, drug-like boronic alcohols (mean QED 0.67, MW 362—the highest drug-likeness of any feature)
- **#429**: Large, flexible boronic amines (mean QED 0.24, MW 605—low drug-likeness due to size)
- **#666**: Halogen-free boronic acid/phenols (flexible, moderate QED)
- **#898**: Brominated, lipophilic boronic acids (mean LogP 4.6—low drug-likeness due to lipophilicity)

This context-dependent specialization mirrors how medicinal chemists evaluate boronic acid-containing molecules: the pharmacological relevance of a boronic acid moiety depends critically on the surrounding molecular scaffold, with factors such as size, halogenation, flexibility, and hydrogen-bonding environment determining whether a boronic acid-containing molecule is a viable drug candidate (e.g., bortezomib) or merely a synthetic intermediate. The SAE's independent encoding of these contexts suggests that SynFlowNet has internalized aspects of this structure–activity reasoning.

0.6 Comparison to Compressive SAE Analysis

Table 7 summarizes the methodological differences between the compressive analysis (Section 2.2) and the extended overcomplete analysis.

Table 7: Comparison of compressive and overcomplete SAE approaches.

	Compressive (§2.2)	**Overcomplete (§5)**
Architecture	$256 \rightarrow 128 \rightarrow 256$	$256 \rightarrow 1024 \rightarrow 256$
Expansion factor	$0.5\times$	$4\times$
Activation	ReLU + ℓ_1	BatchTopK (k=64)
Number of factors/features	128	1024 (470 active)
Sparsity (mean activation)	10.5%	6.25%
Reconstruction (NMSE)	—	0.0008
Probe type	Linear (on factors)	Linear (on features)
Best polarity R^2	0.918	0.915
Best size R^2	0.711	1.000
Best drug-likeness R^2	0.251	0.754
Feature interpretability	Factor–reward correlations	Weight vectors + substructure enrichment

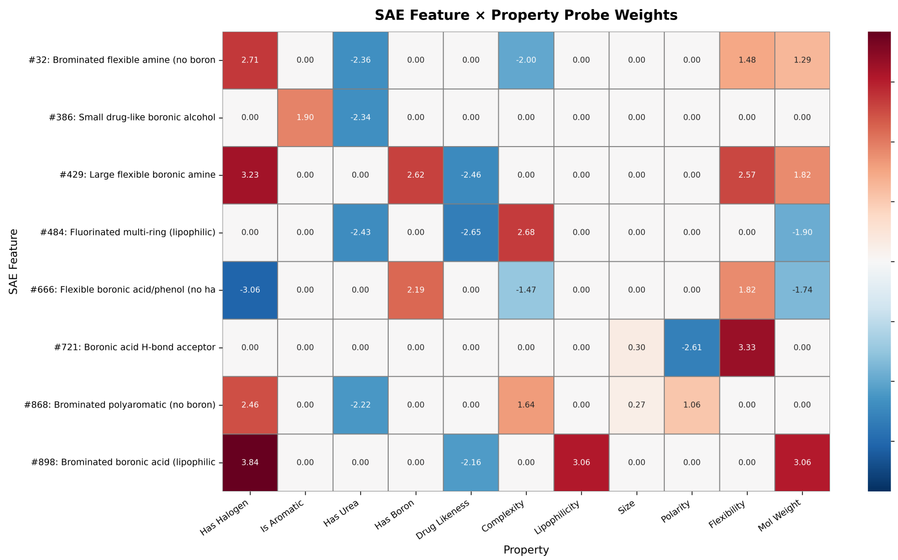

Figure 1: **SAE feature–property weight heatmap.** Rows: eight interpretable SAE features; columns: 11 molecular properties. Cell values are linear probe weights indicating how strongly each SAE feature contributes to each property prediction. Red indicates positive weights (feature activation predicts higher property value); blue indicates negative weights. Monosemantic features appear as rows with a single dominant weight (e.g., #721 for flexibility/polarity), while polysemantic features show weights across multiple columns (e.g., #429). The consistent negative weights in the Drug-Likeness column across Features #484, #429, and #898 reveal the SAE's decomposition of drug-likeness into interpretable structural liabilities.

The overcomplete SAE substantially improves drug-likeness prediction (R^2: $0.251 \rightarrow 0.754$) and size prediction (R^2: $0.711 \rightarrow 1.000$), while maintaining comparable polarity prediction (R^2: $0.918 \rightarrow 0.915$). The overcomplete approach additionally enables feature-level interpretation via probe weight analysis, revealing the specific chemical concepts (halogenation type, scaffold flexibility, boronic acid context) that underlie each property prediction. These results suggest that overcomplete sparse autoencoders are preferable to compressive bottlenecks for mechanistic interpretability of molecular generative models.

A Extended SAE Analysis Details

A.1 SAE Architecture Selection

We explored two SAE families before selecting the final BatchTopK configuration. Table 8 summarizes the hyperparameter sweep.

Table 8: SAE hyperparameter sweep results. ReLU SAEs use ℓ_1 regularization; BatchTopK SAEs enforce exact sparsity via batch-level thresholding.

Type	**Dict Size**	**Param**	**Dead %**	L_0	**NMSE**
ReLU	1024	λ=1e-3	4.1%	629	0.0001
ReLU	1024	λ=1e-2	3.9%	533	0.0002
ReLU	2048	λ=1e-3	10.5%	992	0.0001
ReLU	2048	λ=1e-2	9.7%	863	0.0002
BatchTopK	1024	k=8	94.0%	8	0.0050
BatchTopK	1024	k=16	85.6%	16	0.0026
BatchTopK	2048	k=8	97.6%	8	0.0066
BatchTopK	2048	k=16	94.1%	16	0.0026
BatchTopK+R	**1024**	$\boldsymbol{k}$**=64**	**54%**	**64**	**0.0008**

ReLU SAEs with ℓ_1 coefficients in $[10^{-3}, 10^{-2}]$ achieved excellent reconstruction but failed to produce meaningful sparsity ($L_0 > 500$), behaving as near-identity mappings. BatchTopK with small k (8, 16) enforced exact sparsity but produced 85–97% dead neurons due to the limited training set size. The final configuration—BatchTopK with $k = 64$ and dead neuron resampling (denoted BatchTopK+R)—achieves the best balance: near-perfect reconstruction (NMSE = 0.0008), meaningful sparsity ($L_0 = 64$), and a sufficient number of active features (470) for interpretability analysis.

A.2 Dead Neuron Resampling Schedule

Resampling was applied at epochs 20, 30, 40, and 50, with a 2% cap (approximately 20 features per round). NMSE briefly spiked after each resampling event (0.04–0.06) but recovered within 2–3 epochs, with the model achieving monotonically improving NMSE at convergence: 0.0034 → 0.0022 → 0.0014 → 0.0010 → 0.0008.

A.3 Feature–Property Weight Heatmap

Figure 1 displays the probe weight for each analyzed SAE feature across all 11 molecular properties. Monosemantic features appear as rows with a single strong weight (e.g., Feature #721 for polarity), while polysemantic features show weights across multiple columns (e.g., Feature #429 for halogen, boron, drug-likeness, flexibility, and molecular weight). The sign pattern of weights across properties provides a chemically coherent narrative: features that increase halogen and molecular weight predictions consistently decrease drug-likeness predictions, reflecting the well-known tension between structural elaboration and drug-likeness in medicinal chemistry.

References

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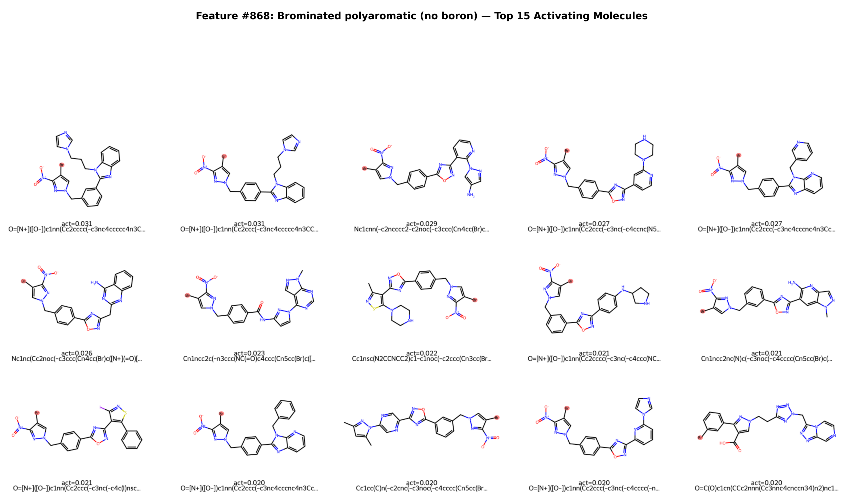
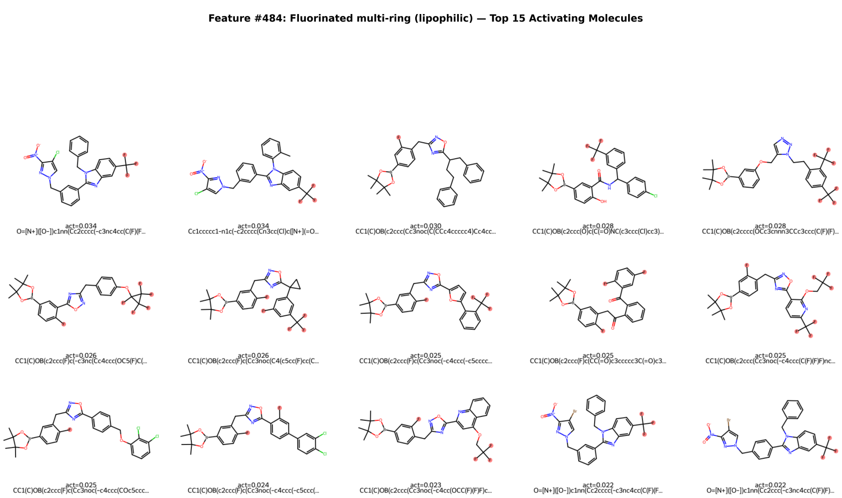


(a) Feature #868: Brominated polyaromatic (no boron). 100% of top-activating molecules contain bromine; boronic acid is absent from all. Br atoms highlighted in red.



(b) Feature #484: Fluorinated multi-ring (lipophilic). 80% of top-activating molecules contain fluorine; mean LogP 5.8. F atoms highlighted in green.